



2026-27

CONCEPTS | CLINICALS | MCQS

BIOCHEMISTRY

- Decode Pathways.
- Crack the Concepts.
- Metabolic cycles made simple
- Vitamins, enzymes & clinical correlations in one glance



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BIOCHEMISTRY

- Fed: Within 2 hours of food intake
- Fasting: 12-18 hours period (lack of food)
- Starvation (prolonged fasting): 1-3 days (lack of food)

Pathways

NEETPG 23, FMGE 24, 23, 21

Anabolic pathways	Catabolic pathway
Glycogenesis	Glycolysis
HMP	Link reaction by PDH complex
Fatty Acid Synthesis Triglyceride Synthesis Cholesterol Synthesis	Glycogenolysis (breakdown of glycogen) β - Oxidation of fatty acids (breakdown of fatty acid) Gluconeogenesis
Enzyme: Lipoprotein Lipase	Enzyme: Hormone sensitive lipase

FMGE 2025, 22

Insulin acts in Fed state, so is always an anabolic hormone

Insulin activates all anabolic pathways –

Exception: Insulin also activates 2 catabolic pathways: 1. Glycolysis 2. Link Reaction

Glucagon activates all catabolic pathways – release during starvation

But it will not activate two catabolic pathways: glycolysis and link reaction

Three pathways occur in both Mitochondria & Cytoplasm (**mnemonic-HUG**)

1. Heme synthesis 2. Urea cycle 3. Gluconeogenesis

DIABETES

FMGE 25

Fasting state / catabolic state → Insulin is absent, GLUT-4 is not active

Hyperglycemia - primary mechanism behind the tissue damage in long standing diabetes

Anabolic pathway inhibited and catabolic pathway activated

In chronic diabetes patient → VLDL synthesis occurs

IN FASTING STATE →

Fatty acid → enter liver for beta oxidation of FA → releases acetyl CoA



1. Enters TCA
2. Converted to ketone body
3. Activate gluconeogenesis

Snowflake cataract- sorbitol deposition



MARKER OF VARIOUS ORGANELLES

FMGE 25

CELL ORGANELLES	MARKER
Plasma Membrane	Na ⁺ K ⁺ ATPase, 5'-Nucleotidase, Adenyl cyclase
Golgi Apparatus	Galactosyl transferase
Endoplasmic reticulum	Glucose-6-Phosphatase
Peroxisomes	Catalase
Lysosome	Acid phosphatase

FUELS FOR BODY IN VARIOUS STATES

NEETPG 25

organs	fed	fasting	starvation
brain	G	G	KB
heart	FA	FA	KB
liver	G	FA	AA
muscle	G	FA	FA and KETONE
RBC	G	G	G

G- glucose KB- ketone bodies AA- amino acid FA- fatty acids

• **Acetyl CoA only activates the first step of gluconeogenesis but can never be converted to glucose. It is never glucogenic.**

FMGE 24

Galactose	Glucose + lactose
Lactulose	Galactose + fructose
Maltose	Glucose + glucose
Lactose	Galactose + glucose
Sucrose	Glucose + fructose

GLUCOSE TRANSPORTER**NEETPG 24, FMGE 24**

Facilitative Bidirectional Transporters:	Location	Functions
GLUT-1	Brain kidney, colon, placenta, RBS	Glucose uptake
GLUT-2	Liver, Pancreatic B-cell, Small Intestine, kidney	Rapid uptake or release of glucose
GLUT-3	Brains, Kidney, Placenta	Glucose uptake
GLUT-4	Heart, skeletal Muscle, adipose tissue	Insulin – Stimulated glucose uptake
GLUT-5	Heart, skeletal Muscle, adipose tissue	Uptake of fructose
SGLT-1	Small Intestine & kidney	Active uptake of glucose against concentration gradient
SGLT-2	kidney	

GLYCOLYSIS**GAIN OF ATP**

- (1) Aerobic glycolysis produces NET=7 moles of ATP per mole of glucose
 (2) Anaerobic glycolysis produces NET= 2 moles of ATP per mole of glucose

IN RBC- 2 ATP

Rate limiting enzyme- **phosphofructokinase-1**

Pathways producing zero ATP

• HMP • Uronic acid pathway • RL shunt • α -oxidation of FA**Inhibitors of glycolysis-**

Iodoacetate	G3PD
Arsenic	G3PG
NaF	Enolase
Oxamate	LDH

FMGE 25

Histidine → AA → act as buffer in blood due to its ability to accept & donate protons at physiological pH

Note-

Pyruvate kinase deficiency is the most common enzyme defect known in glycolysis.
Pyruvate kinase deficiency leads to hemolytic anemia

Nucleotides -

Purines (A,G)—2 rings.

Pure As Gold.

Pyrimidines (C,U,T)—1 ring

CUT the pyramid.

Chargaff's Rules = number of purines = number of pyrimidines

Adenine = Thymine

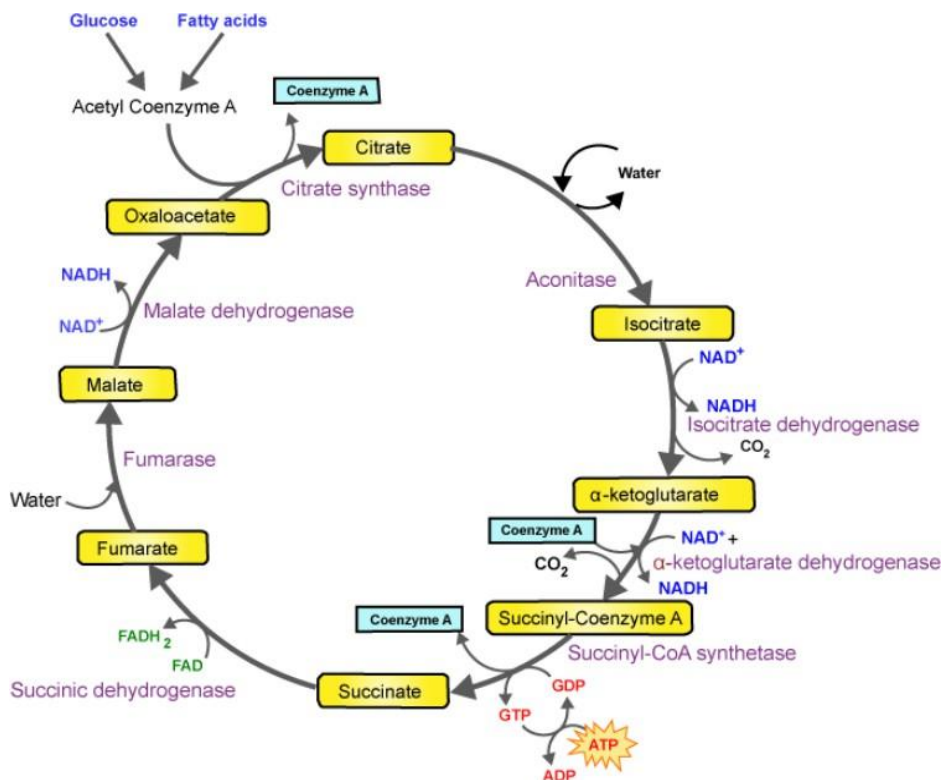
Guanine = Cytosine

Purines = Pyrimidines

CITRIC ACID CYCLE (KREB CYCLE OR TCA CYCLE)

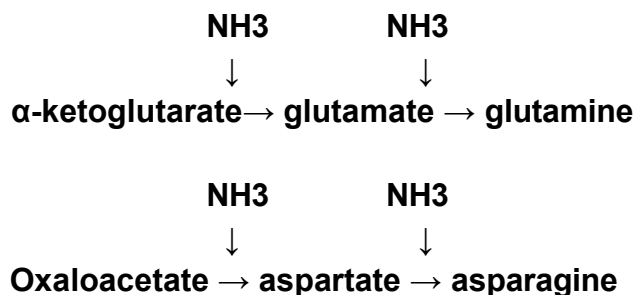
Location: Mitochondrial matrix

Total 10 ATP are produced per turn of citric acid cycle



Imp points

FMGE 23



Malate (4C): Intermediate of TCA Cycle

Malonate (3C): Inhibitor of TCA Cycle

Malonate / Malonyl CoA is an inhibitor of →

1. TCA Cycle [Enzyme inhibitor is Succinate Dehydrogenase]
2. ETC [component inhibited is Complex II]
3. Beta Oxidation of Fatty Acids (Enzyme inhibited is Carnitine Palmitoyl Transferase-1 {CPT-1})

Glycolysis-

Three Regulatory /Irreversible Enzymes

1. Hexokinase /Glucokinase
2. Phosphofructokinase-1 (PFK-1)
3. Pyruvate Kinase

Gluconeogenesis

FMGE 25

3 Rate Limiting Enzymes

1. Pyruvate carboxylase
2. PEPCK (PhosphoenolPyruvate carboxykinase)
3. Fructose 1-6 Bisphosphatase (Most Important)

TCA Cycle / Krebs's Cycle

3 Rate Limiting Enzymes

1. Citrate synthase
2. α -Ketoglutarate dehydrogenase
3. Isocitrate dehydrogenase: Most important → Stimulated By ADP → Inhibited by ATP and NADH

Vitamins play role in TCA cycle- B1,B2,B3 and lipoic acid

ETC INHIBITORS-

NEETPG 24,22,FMGE 23,22

Complex I (NADPH CoQ reductase) - rotenone,phenobarbitone

Complex II (succinate CoQ reductase) -malonate

Complex III (cytochrome C reductase)-phenformin,Antimycin A

Complex IV (cytochrome C oxidase) -cyanide,CO,H₂S

ATP synthase - oligomycin

Uncouplers

Drugs - dinitrophenol ,Natural- thermogenin ,thyroxine

Cofactors-

Kinases, phosphorylase,carboxylase- use manganese as cofactor

Pyruvate kinase require- K⁺

NAD need vit B3 FAD need vit B2 CoA need vit B5

Cu required for all oxidases

Exception → Mitochondrial SOD requires Manganese

→ Xanthine oxidase and Sulphite oxidase do not require copper, they require Molybdenum (Mb)

Pyruvate dehydrogenase complex

FMGE 24,22

Contains 3 enzymes requiring 5 cofactors:

1. Thiamine pyrophosphate (B1)
2. Lipoic acid
3. CoA (B5, pantothenic acid)
4. FAD (B2, riboflavin)
5. NAD⁺ (B3, niacin)

Deficiency causes -buildup of pyruvate that gets shunted to lactate (via LDH) and alanine (via ALT).

NEETPG 23

Pathway	Rate limiting enzyme	Activators	Inhibitors
Glycogenesis	Glycogen synthase	Insulin, glucose 6-P	Glucagon, Epinephrine
Glycolysis	PFK-1	AMP, fructose 2,6 bisphosphate	ATP, Citrate
Glycogenolysis	glycogen phosphorylase	cAMP, kinase, Glucagon, Epinephrine	Insulin, Phosphatase, ATP, Glucose, Glucose-6-P, Fructose-1 P

NEETPG 22, FMGE 24,21

Pathway	RLE
ketone body synthesis	HMG Co-A synthase
Cholesterol body synthesis	HMG CoA reductase
Fatty acid synthesis	Acetyl CoA carboxylase or malonyl CoA synthetase
TCA cycle	Isocitrate dehydrogenase
HMP shunt	Glucose-6-phosphate dehydrogenase (G6PD)
De novo pyrimidine synthesis	Carbamoyl phosphate synthetase II
De novo purine synthesis	Glutamine-phosphoribosylpyrophosphate (PRPP) amidotransferase
Urea cycle	Carbamoyl phosphate synthetase I

GLYCOGEN STORAGE DISEASES**NEETPG 24,23, FMGE 25,24,21,26**

glycogenesis	Name	Deficiency and symptoms
Type 1	Von Gierke's disease	Glu-6-phosphatase ↑↑ Glycogen in liver , ↑ blood lactate , ↑triglycerides, Doll like facies
Type II	Pompe's disease	Lysosomal α-1,4 and α-1, 6 glucosidase Cardiomyopathy , hypotonia, enlarged tongue, and systemic findings lead to early death.
Type III	Cori's disease	Absence of debranching enzyme normal blood lactate levels, Limit dextrin –like structures accumulate in cytosol
Type IV	Andersen's disease	Absence of branching enzyme failure to thrive in early infancy.
Type V	McArdle's syndrome	Absence of muscle phosphorylase ↑↑ glycogen in muscle, but muscle cannot break it down painful muscle cramps
Type VI	Her's disease	liver phosphorylase

Mnemonic- ABCD

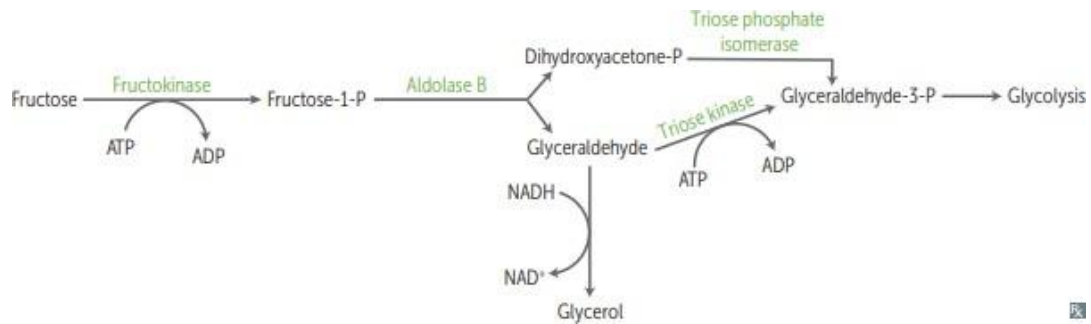
A- anderson disease B- branching enzyme

C- cori's disease D- debranching enzyme

McArdle = muscle

FRUCTOSE METABOLISM

Essential fructosuria	Hereditary fructose intolerance
Enzyme deficiency - Fructokinase	Enzyme deficiency - Aldolase B
Fructose is not trapped in cells; hexokinase converts it to fructose-6-phosphate.	Fructose-1-phosphate buildup decreases available phosphate, inhibiting glycogenolysis and gluconeogenesis.
Asymptomatic, benign. Fructose appears in blood and urine	hypoglycemia, jaundice, cirrhosis, vomiting appear after eating fruit, juice, or honey



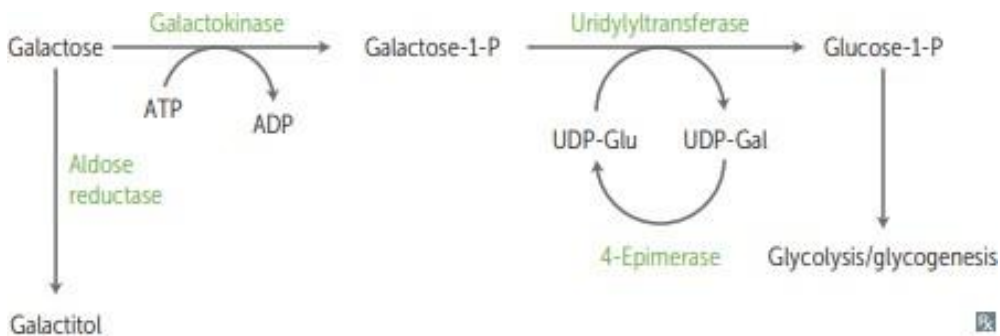
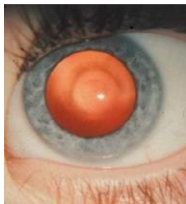
GALACTOSE METABOLISM

NEETPG 24, FMGE 25, 24, 23, 21

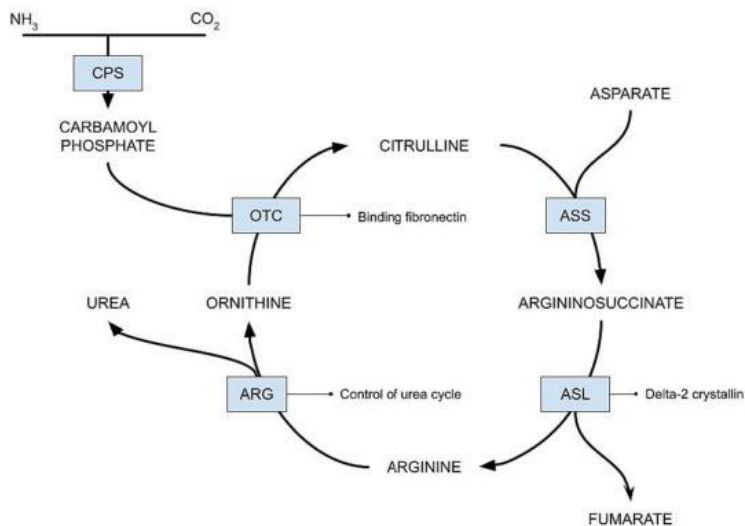
Galactosaemia condition is characterised by inability to metabolise dietary galactose due to deficiency of either of these enzymes

- Galactokinase
- galactose-1-phosphate uridyl transferase

Oil drop cataract



UREA CYCLE



CPS-I (Carbamoyl phosphate synthetase - I) – RLE

FMGE 25

1. OTC (Ornithine TransCarbamoylase)

Absent in brain

OTC deficiency is most common urea cycle defect

2. Arginase

Absent in kidneys

So, End-product of urea cycle in kidneys: Arginine

Urea cycle disorders

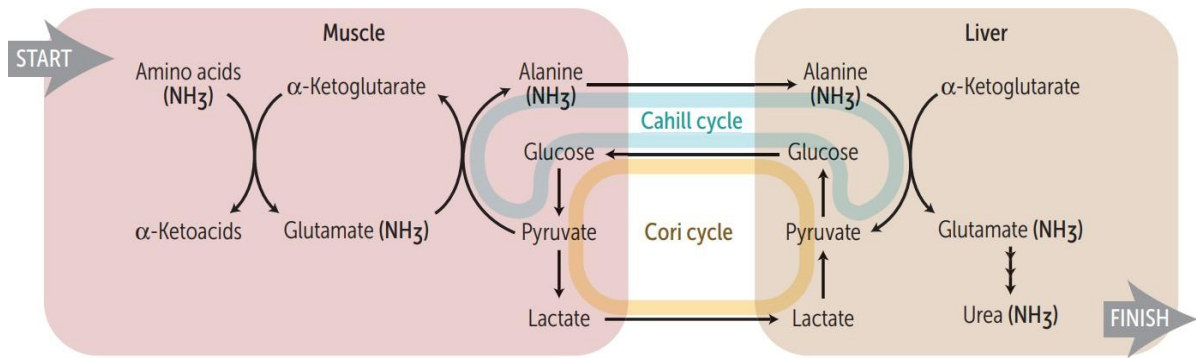
FMGE 26

Disease	Enzyme deficient	Substrate accumulated
Hyperammonemia Type I	CPS – I	NH ₃ (severe hyperammonemia , high glutamine ,low citrulline
Hyperammonemia Type II (MC UCD)	Ornithine transcarbamoylase (OTC)	NH ₃ , OMP, UMP, Orotic acid
Citrullinemia type I	Arginosuccinate synthetase	NH ₃ , Citrulline
Hyperargininemia	Arginase	NH ₃ , Arginine

Urea cycle transporter defects

Disease	Enzyme deficient	Substrate accumulated
Citrullinemia type II	Citrin (Aspartate-Glutamate Transporter)	Citrulline
HHH Syndrome	Ornithine transporter	Hyperammonemia Hyperornithinemia Homocitrullinuria

Transport of ammonia by alanine



FMGE 24

Heme derived from glycine

Ammonia transported from blood to liver in glutamine form

The main amino acids used in purine synthesis are glycine, aspartate, and glutamine

Essential fatty acid

- Linoleic acid • α -linolenic acid
- Semi Essential fatty acid: Arachidonic acid

Sphingolipidoses

NEETPG 23,22,FMGE 24,21,26

DISEASE	FINDINGS	DEFICIENT ENZYME
Tay-Sachs disease	Mental retardation, blindness, muscular weakness “cherry-red” spot on macula	Hexosaminidase A
Fabry disease	Early: triad of episodic peripheral neuropathy, angiokeratoma	α-galactosidase A
Krabbe disease	Peripheral neuropathy, destruction of oligodendrocytes	Galactocerebrosidase
Gaucher disease*	Hepatosplenomegaly, pancytopenia, osteoporosis lipid-laden macrophages resembling crumpled tissue paper	Glucocerebrosidase
Niemann-Pick	Progressive neurodegeneration,	Sphingomyelinase.

disease(AR)*	hepatosplenomegaly “ cherry-red ” spot on macula , foam cell	
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Garrod’s tetrad: 4 diseases (MN- CAAP)

FMGE 2025

1. C → Cystinuria
2. A → Alkaptonuria
3. A → Albinism
4. P → Pentosuria (Essential) → L-xylulose excreted in urine of patient

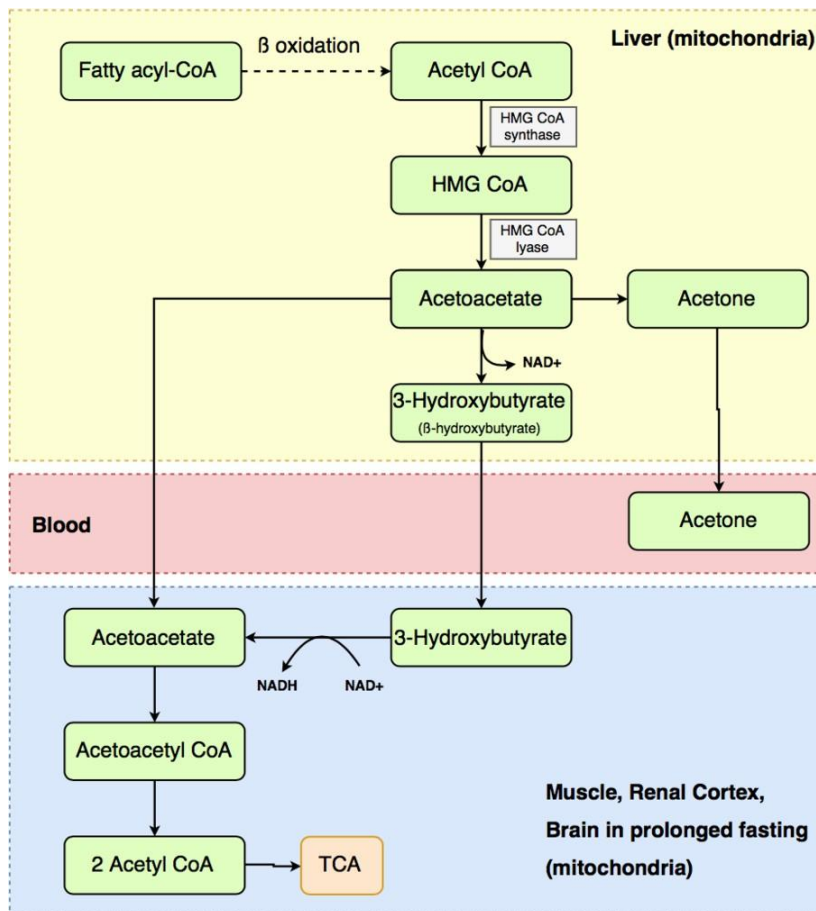
Mucopolysaccharidoses

NEET PG 25

Disease	Findings	Deficient enzyme	Accumulated substrate
Hurler syndrome	Developmental delay, hirsutism, skeletal anomalies, clouded cornea	α -L-iduronidase	Heparin sulfate Dermatan sulfate
Hunter syndrome	No corneal clouding Trick - Hunters see clearly and aggressively aim for the X (x-linked recessive)	Iduronidase -2 sulfatase	Heparan sulfate Dermatan sulfate
Maroteaux Lamy Syndrome	No mental retardation	Aryl Sulfatase B	Dermatan sulfate

Ketogenesis

Occurs in liver



The liver cannot use ketone bodies as it lacks thiophorase.

Tests for Detection of Ketone Bodies

1. Rothera's test $\rightarrow$ given positive by Acetoacetate & Acetone
2. Gerhard's test $\rightarrow$ given positive by Acetoacetate only

FMGE 24

Zellweger Syndrome $\rightarrow$ Defect in peroxisomes in all body cells causing defect in oxidation of VLCFA

FMGE 25

MCAD deficiency: Inability to break down medium-chain fatty acids $\rightarrow$ causes hypoketotic hypoglycemia, vomiting, lethargy, and seizures, especially during fasting.

FMGE 26

Refsum's disease : increase phytanic acid in peroxysomes, defect in α oxidation,

AMINO ACID TRANSPORTER DEFECTS

Hartnup's Disease-

FMGE 24

Common transporter for tryptophan and neutral amino acids

– Patient have pellagra i.e. Vit-B3 deficiency as tryptophan has role in forming Vit-B3 (60 mg Tryptophan makes 1 mg Niacin) – Serum tryptophan levels not raised

Apolipoproteins

1. Apoproteins A:

- These include Apo A-1 and Apo A-II. • They are the major apoproteins of HDL

2. Apoproteins B:

- These include ApoB-48 of chylomicrons and ApoB-100 of LDL & VLDL.
- B-48 synthesised in the intestine and B-100 in the liver.

3. Apoproteins C:

- They include Apo C-1, Apo C-II and Apo C-III.
- Apo C-I is present in VLDL & HDL whereas C-II and C-III are present in VLDL, HDL and chylomicrons.
- Apo A-1 is an activator of LCAT & Apo C-II is activator of lipoprotein lipase.

Neurodegenerative disorder = Apo D

Hyperlipoproteinemias ,26

NEETPG 25,24,FMGE 25

Type	Classification	BLOOD LEVEL	Clinical findings
I	Lipoprotein lipase deficiency	Chylomicrons, TG	Pancreatitis, hepatosplenomegaly, and eruptive/pruritic xanthomas
Ila	Hypercholesterolemia	LDL	Accelerated atherosclerosis (may have MI before age 20), tendon (Achilles) xanthomas, and corneal arcus.
Ilb	Combined hyperlipidemia	LDL, VLDL	Accelerated atherosclerosis (may have MI before age 20), tendon (Achilles) xanthomas, and corneal arcus.
III	Dysbetalipoproteinemia	Chylomicrons, VLDL	Premature atherosclerosis, tuberoeruptive and palmar xanthomas
IV	Hypertriglyceridemia	VLDL	Hypertriglyceridemia (> 1000 mg/dL) can cause acute pancreatitis.

WOLMAN'S DISEASE

Lysosomal storage disease

Enzyme deficient is Acid Lipase

↑↑ **Ch esters and TG**



C/f –

Watery green diarrhoea

Relentless vomiting and failure to thrive

Calcification of adrenals is pathognomonic feature

PUFAs

Omega 3	Omega 6
1. Cervonic acid/DHA (DocosaHexaenoic Acid) 2. α -Linolenic acid 3. Timnodonic acid/EPA (EicosaPentaenoic Acid)	1. γ -Linolenic acid 2. Linoleic acid 3. Arachidonic acid

Lipoproteins

Density $\propto$ 1/Size of LP

Density $\propto$ 1/TG content

Density $\propto$ % of protein

Density -

Chylomicron < Chylomicron remnant < VLDL remnant/IDL < LDL < HDL

Composition of lipoproteins

Lipoprotein	Lipid present	Protein/Apoprotein present
Chylomicron	TG	Apo B48
Chylomicron Remnant	TG + chol	Apo B48 + Apo E
VLDL	TG (endo)	Apo B100
VLDL Remnant (IDL)	TG + Chol	Apo B100 + Apo E
LDL	Chol	Apo B100 + Apo E
HDL	Chol-ester	Apo A, Apo C and Apo E

Tangier's disease.

NEETPG 25

Mutation in ABCA-1 gene

Decreased HDL

Cholesterol accumulation in various body tissues

Like in tonsils → Large orange/yellow tonsils → characteristic feature



LCAT Deficiency

Complete deficiency	Partial deficiency
Norum's Disease Severe hemolytic anemia and End Stage Renal Disease (ESRD)	Fish-eye Disease C/f - are same as Norum's disease but No hemolytic anemia or ESRD

Metabolic Defects

1. Maple syrup Urine Disease-

FMGE 21

- Blocked degradation of branched amino acids (Isoleucine, Leucine, Valine) due to
- branched chain α -ketoacid dehydrogenase (B1)
- urine smells like maple syrup/burnt sugar

2. Phenylketonuria

- ↓ phenylalanine hydroxylase (PAH)
- intellectual disability, microcephaly, seizures, hypopigmented skin, eczema, musty body odor (due to phenylacetate), Severe MR (due to Phenylalanine)
- Best test → Mass spectrometry (MS) (best test for any inborn error of metabolism)

3. Alkaptonuria

NEETPG 23, FMGE 25,24,22

- Congenital deficiency of homogentisate oxidase
- pigment-forming homogentisic acid builds up in tissue
- bluish-black connective tissue, ear cartilage, and sclerae (ochronosis), urine turns black on prolonged exposure to air



4. Homocystinuria

FMGE 23

1. Acquired • Due to vitamin B6, B9 and B12 deficiency
2. Genetic • Due to Enzyme Cystathionine beta Synthase defect
Ectopia lentis- downward



PTM of Collagen requires: Three enzymes

NEETPG 25, FMGE 25

1. Prolyl hydroxylase 2. Lysyl hydroxylase 3. Lysyl oxidase

PORPHYRIAS

FMGE 26

Types of porphyrias	Enzyme defect	Characteristics
Acute intermittent porphyrias	Uroporphyrinogen I synthase	Abdominal pain Neuropsychiatric symptoms
Porphyria cutanea tarda	Uroporphyrinogen decarboxylase *	Photosensetivity

LDH ISOENZYMES

In normal persons: LDH 2 > LDH 1

In case of Myocardial Infraction: LDH 1 > LDH 2

Glycosaminoglycans

FMGE 25

Hyaluronic acid- role in wound healing, synovial fluid, vitreous humor

Keratin sulfate- cornea (corneal transparency provided by keratin)

Dermatan Sulfate- Found in Skin, Blood vessels, heart valves

Heparan Sulfate- Has a role in cell-cell adhesion e.g. retinal cell-cell attachment

Types of collagen

Type I- Most common (90%)—Bone (made by osteoblasts), Skin

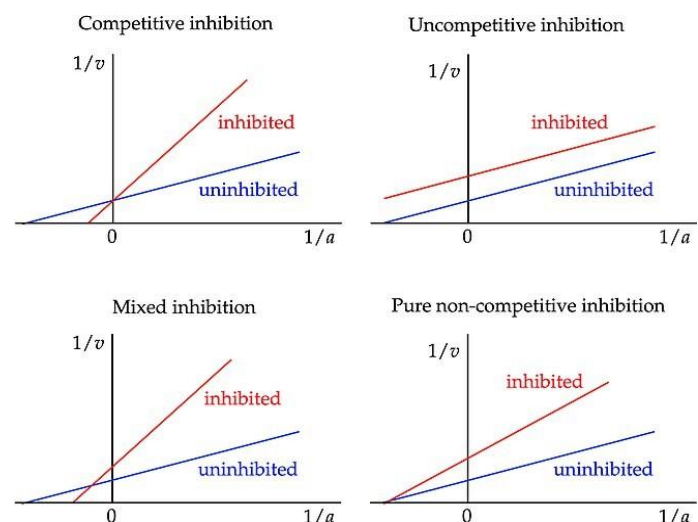
Type II- cartilage

Type III- blood vessels eg- Ehlers Danlos syndrome

Type IV- basement membrane. Defective in Alport syndrome

Lineweaver Burk plot for inhibition

	K _m	V _{max}
competitive	↑	same
Non competitive	same	↓
Un competitive	↓	↓



Blotting procedure

SNoW-DRoP:

Southern = DNA Northern = RNA Western = Protein

-21st AA → Selenocysteine → Given by UGA

-22nd AA → Pyrrolysine → Given by UAG

-UGA and UAG are stop codons and usually do not code for any amino acid, but by Co-translational modification, they make Selenocysteine and Pyrrolysine

VITAMINS

Fat soluble – vit A, D, E, K

Water soluble – vit B, C

Vitamin A

NEETPG 25,24,22,FMGE 24,23

Active form: • RETINOL • RETINAL • RETINOIC ACID

It is a teratogenic

Functions of vitamin A

- Retinal - vision
- Retinol - reproductive system
- β -Carotene - antioxidant property

Deficiency

Phrynoderma

Night blindness, - Nyctalopia- Xerophthalmia-Bitot's spots, Keratomalacia

Most specific manifestation- bitot's spots

Earliest symptoms- night blindness

WHO CLASSIFICATION	OF XEROPHTHALMIA
XN	Night blindness (nyctalopia)
X1A	Conjunctival Xerosis
X1B	Bitot's spots
X2	Corneal Xerosis
X3A	Corneal ulceration (<1/3 rd)
X3B	Corneal ulceration(>1/3 rd)
XS	Corneal scar
XF	Xerophthalmic fundi

Excess

Acute toxicity—nausea, vomiting, ↑ ICP (eg, vertigo, blurred vision).

Chronic toxicity—alopecia, dry skin (eg, scaliness), and idiopathic intracranial hypertension

Vitamin D

FMGE 21

Active form: Calcitriol or 1,25 dihydroxy cholecalciferol

FUNCTION

-intestinal absorption of Ca^{2+} and PO_4^{3-} —bone mineralization at low levels, bone resorption at higher levels.

REGULATION - ↑PTH, ↓ Ca^{2+} , ↓ PO_4^{3-}

Deficiency in Children –

Rickets • Beaded appearance of RIBs aka Rachitic Rosary • Hot cross bun appearance of head due to nonclosure of fontanelle • Bowlegs (genu varum) and Knock Knees

Deficiency in adults –

Osteomalacia • Soft bones with frequent fracture • Waddling gait

Hypervitaminosis D

Symptoms → Hypercalcemia • Hypercalciuria causing Renal stones • Calcification of Soft tissues: calcium deposited in blood vessels causing hypertension

VITAMIN K

NEETPG 23, FMGE 23

Factors dependent on vitamin K are II, VII, IX, & X

FUNCTION

Activated by epoxide reductase to the reduced form, which is a cofactor for the γ -carboxylation of glutamic acid residues.

Warfarin inhibits vitamin K–dependent synthesis of these factors and proteins

C/F OF DEFICIENCY: Hemorrhagic disease of newborn Prolong BT, CT, PT

Note- breast milk lacks vitamin D and K

THIAMINE: (vitamin B1).

NEETPG 25,23,22,FMGE

25,24,21,26

Enzymes that need TPP- triple pyrophosphate (“Be APT”)

B – Branched-chain ketoacid dehydrogenase

A – α -Ketoglutarate dehydrogenase

P – Pyruvate dehydrogenase

T – Transketolase

Assessment of deficiency: by Erythrocyte **T**ransketolase estimation.

Deficiency Syndromes-

1. Wernicke encephalopathy - Acute, reversible, life-threatening neurological condition. Symptoms: Confusion, Ophthalmoplegia/ Nystagmus, Ataxia (CorONA beer) (**FMGE 26**)

2. Korsakoff syndrome- Amnestic disorder due to chronic alcohol overuse; presents with confabulation, personality changes, memory loss (permanent)

3. Wet beriberi- High-output cardiac failure (due to systemic vasodilation)

RIBOFLAVIN (VIT. B2)

FMGE 21

Active form: FAD, FMN

DEFICIENCY

Cheilosis (inflammation of lips, scaling and fissures at the corners of the mouth), “magenta” tongue, corneal vascularization.



Glossitis



Cheilosis



Itching and burning eyes

Deficiency assessment done by: RBC glutathione reductase activity

NIACIN Vit B3.

NEETPG 24,23,FMGE 24,22,21

Coenzyme forms – NAD & NADP

Derived from tryptophan

Synthesis requires vitamins B2 and B6

Used to treat dyslipidemia (VLDL, HDL)

Deficiency leads to Pellagra

In carcinoid syndrome & hartnup's disease 60% of tryptophan is catabolized via serotonin pathway and causes pellagra.

Clinical symptoms - Dermatitis, Diarrhoea, Dementia, & Death (4 D's)



Pellagra- vitamin B3 less

Podagra = vitamin B3 OD (overdose).

Vitamin B5

Component of coenzyme A

deficiency- Dermatitis, enteritis, alopecia, adrenal insufficiency may lead to burning sensation of feet ("burning feet syndrome"; distal paresthesias, dysesthesia)

Vitamin B6.

NEETPG 24

Deficiency -Convulsions, hyperirritability, peripheral neuropathy (deficiency inducible by isoniazid and oral contraceptives)

Vitamin B7 (Biotin)

NEETPG 23

Cofactor for Carboxylation Enzymes

Main Enzymes

1. Pyruvate carboxylase – in gluconeogenesis

→ Pyruvate (3C) → Oxaloacetate (4C)

2. Acetyl-CoA carboxylase – in fatty acid synthesis

→ Acetyl-CoA (2C) → Malonyl-CoA (3C)

3. Propionyl-CoA carboxylase – in fatty acid oxidation & amino acid breakdown

→ Propionyl-CoA (3C) → Methylmalonyl-CoA (4C)

Deficiency -Dermatitis, enteritis, alopecia. Caused by long-term antibiotic use or excessive ingestion of raw egg whites

Vitamin B9

Deficiency – causes Megaloblastic anaemia.

Folate absorbed in jejunum

Important for the synthesis of nitrogenous bases in DNA and RNA.

DEFICIENCY

Macrocytic, megaloblastic anaemia; hypersegmented polymorphonuclear cells (PMNs); glossitis; no neurologic symptoms (as opposed to vitamin B12 deficiency).

Labs: homocysteine, normal methylmalonic acid levels

Deficiency can be caused by several drugs (eg, phenytoin, trimethoprim, methotrexate)

Vitamin B12

NEETPG 25, FMGE 24,21

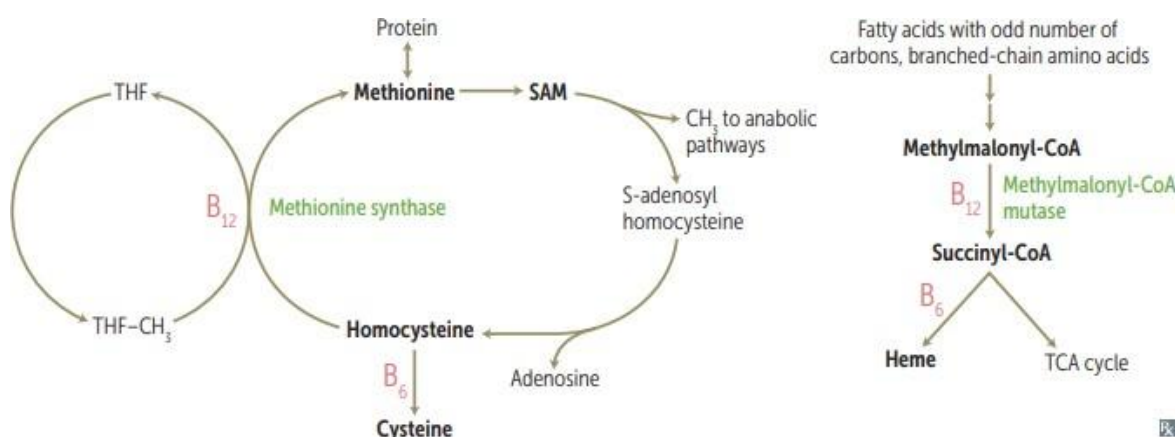
Important for DNA synthesis

Absorb in ileum and Absorption is with intrinsic factor

DEFICIENCY

Macrocytic, megaloblastic anemia; hypersegmented PMNs; paresthesias and subacute combined degeneration (degeneration of dorsal columns) due to abnormal myelin.

Associated with ↑ serum homocysteine and methylmalonic acid levels, along with 2° folate deficiency.



Universal methyl group donor - S-methionine

Vitamin C

NEETPG 24,23, FMGE 25,21

Helps in absorption of iron from the intestine.

converts dopamine to NE

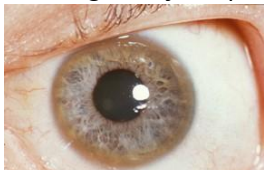
Deficiency may be precipitated by a tea and toast diet.

Vitamin C deficiency causes scurvy

Scurvy—swollen gums, easy bruising, petechiae, hemarthrosis, anemia, poor wound healing, perifollicular and subperiosteal hemorrhages, “corkscrew” hair



NEETPG 24,FMGE 21

MENKE DISEASE	WILSON DISEASE
Copper deficient	Copper toxicity
ATP 7A defect	ATP 7B defect
Ceruloplasmin ↓	Ceruloplasmin ↓
Grey pigmented hair, brittle kinky hair 	KF ring in eyes (descemet's membrane) 

Zinc deficiency

NEETPG 23,FMGE 25,24

Clinical features -

Delayed wound healing

Male hypogonadism

Adult hair (axillary, facial, pubic)

Associated with acrodermatitis enteropathica

Selenium deficiency

Keshan's Disease

Clinical Features →

Cardiomyopathy

Muscular weakness and muscular dystrophy

Inability to gain weight

Eczema

Selenium - cofactor of glutathione peroxidase

VACUTAINER TUBES

Color	Anticoagulant	Uses
	No anticoagulant	Serological examination in biochemistry
	Sodium fluoride	Glucose estimation

	EDTA	ESR,complete hemogram
	3.2% sodium citrate	Coagulation studies like PT,aPTT
	EDTA	ABO grouping

FMGE 26

- **Steroid hormone** : intracellular receptor , DNA motif that enables steroids hormone receptors to attach to their target sites on DNA is the **zinc finger motif**
- ***DNA FINGERPRINTING** : short tendon repeats are used as a genetic marker in dna fingerprinting

DNA DAMGE REPAIR

DNA DAMGE REPAIR TYPE	DISEASE
NHEJ (non homologus end joining repair)	SCID
DEFECTIVE HOMOLOGUS REPAIR	BLOOM SYNDROME WERNER SYNDROME BREAST CANECR
NER (nucleotide excision repair)	XERODERMA PIGMENTOSUM
BER (base excision repair)	MUTYH ASSOCIATED POLYPOSIS
MMR (mismatch repair)	HNPCC

RNA EDITING : apo B48 synthesis is involve in rna editing

